

Earth-Agent-Pro: Towards Real-World Full-Chain Earth Observation with Agents

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Abstract—Real-world Earth observation (EO) analysis requires agents to be capable of translating high-level scientific questions into executable workflows for acquiring observations, preparing data, performing domain computations, and deriving conclusions from runtime evidence. However, existing EO agents and benchmarks generally cover only part of this chain: agents typically start from supplied observations, whereas benchmarks typically provide prepared inputs or candidate answers, leaving full-chain open-world EO execution largely untested. We present Earth-Agent-Pro, an execution-adaptive Plan-and-Execute framework that uses expert-authored skills to constrain workflow planning and runtime tool use. It records planned steps, accepted evidence, and their dependency links in workflow-centered structured memory, then repairs only the affected workflow suffix when runtime evidence invalidates a step. Role-specific training adapts the underlying large language model (LLM) through separate planner and executor adapters: sequence-level supervised fine-tuning targets planner workflow composition, while node-level group relative policy optimization with locally verifiable rewards targets executor tool-argument grounding. To evaluate full-chain open-world EO execution, we introduce Earth-Bench-Pro, a 744-question EO agent benchmark that instantiates 248 expert-curated task cores under three matched regimes. Across RGB imagery, spectral observations, and remote sensing products, its 248 Open-World Execution questions pair high-level requests with runtime data requirements, executable trajectories, and open-ended answers grounded in execution evidence. Under a shared GPT-5 backbone, Earth-Agent-Pro reaches 66.13% LLM-as-Judge accuracy, outperforming ReAct by 20.95 points in LLM-as-Judge accuracy and 24.44 points in Tools-In-Order. Jointly tuning the Planner and Executor adapters raises Qwen3.5-9B LLM-as-Judge accuracy from 38.31% to 50.00%, an 11.69-point gain over the untuned configuration. Planning-only evaluation and execution with the reference workflow further show that the adapters improve workflow composition and argument grounding, respectively. In conclusion, Earth-Bench-Pro and Earth-Agent-Pro support evaluation and training of EO agents from user requests to evidence-grounded scientific answers. The code and datasets of this work will be released soon.

Index Terms—Earth observation, remote sensing agents, open-world execution

I. INTRODUCTION

Earth observation (EO) turns satellite and aerial measurements into quantitative evidence for agricultural monitoring, disaster assessment, urban analysis, and environmental change research [1]–[15]. Producing such evidence from an EO question remains expert-intensive [16]–[18]. An EO analysis begins

by matching the target phenomenon, region, and period to suitable observations. Analysts then choose the data source, product, bands, and sampling parameters before acquiring and preprocessing the data. The prepared observations enter a domain-specific computation pipeline that yields the requested scientific quantities. Completing this workflow requires expertise in sensors, data products, geospatial processing, and domain science, which raises the entry barrier for users outside the EO community and limits access to reproducible analysis.

Recent work has expanded EO capabilities beyond interpreting supplied observations to include multistep analysis execution. Vision-language models interpret supplied images and image sequences [19]–[21], while tool-augmented agents select operations, execute code, and construct multistep workflows [22]–[26]. Benchmarks now span fixed-input question answering, multistep geographic information system (GIS) tool use, and open-environment pipeline construction [27]–[30]. Together, these methods and benchmarks cover more of the request-to-answer chain, but evaluations differ in their scientific tasks, input assumptions, and answer formats. Measuring progress toward autonomy therefore requires a matched evaluation that removes expert scaffolding while keeping the scientific objective fixed.

Our prior Earth-Agent and Earth-Bench provide the task and evaluation foundation for this comparison [31]. Earth-Agent connects a language model to EO tools through the Model Context Protocol, while Earth-Bench evaluates final answers and reference trajectories across red-green-blue (RGB) imagery, spectral observations, and remote sensing products. This pairing evaluates both the scientific result and the execution path. Earth-Bench, however, begins with expert-prepared local inputs and multiple-choice questions, leaving runtime data discovery and open-ended answer generation outside its evaluation boundary. We build on this foundation by preserving each of Earth-Bench’s 248 scientific objectives as a fixed task core and moving runtime data discovery, acquisition, and open-ended answer generation into the evaluated workflow.

The resulting request-to-answer setting, which we call *open-world EO execution*, starts from a request that specifies a phenomenon, region, and period. The agent must locate runtime inputs, acquire suitable observations when required, prepare them, perform domain computations, and derive an answer from runtime evidence. Files, intermediate artifacts, and tool results appear only during execution, so later op-

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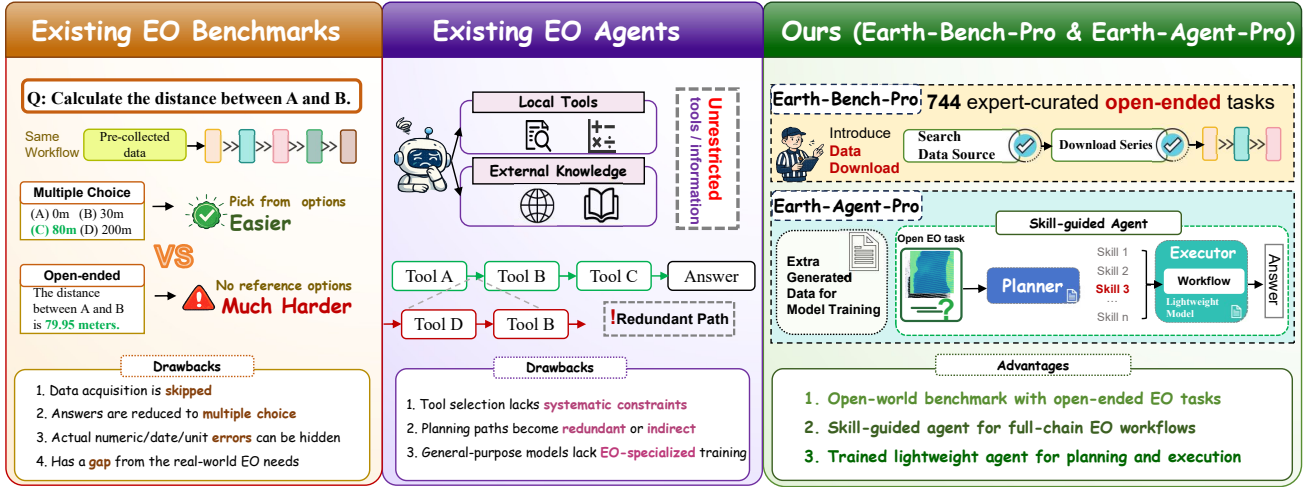


Fig. 1. **Overview of our work.** Existing EO benchmarks rely on pre-collected data and multiple-choice evaluation, while existing EO agents suffer from unconstrained tool use and redundant planning. Our work introduces an open-ended EO benchmark (Earth-Bench-Pro) and a trained skill-guided agent (Earth-Agent-Pro) for full-chain EO task planning and execution.

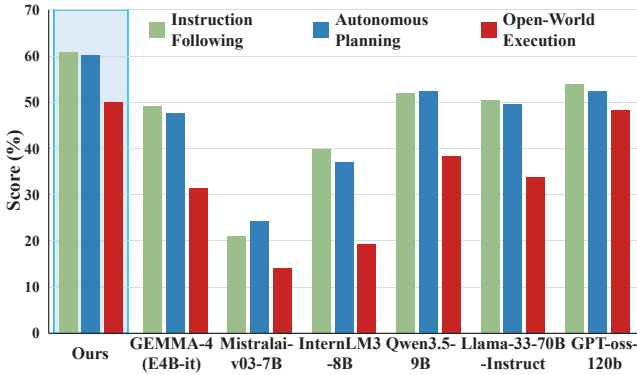


Fig. 2. LLM-as-Judge accuracy across Earth-Bench-Pro's three regimes. **Ours** is our role-specialized Qwen3.5-9B.

erations depend on accepted upstream evidence. The agent must track workflow dependencies, ground each operation in the evolving state, and revise affected future operations when evidence invalidates the current plan.

We develop **Earth-Bench-Pro** and **Earth-Agent-Pro** as a paired benchmark and agent for open-world EO execution. Earth-Bench-Pro holds each scientific objective fixed across matched regimes that progressively remove the reference workflow and prepared inputs. Earth-Agent-Pro separates workflow planning from runtime operation grounding and constrains both stages with expert-authored skills. Workflow-centered structured memory links pending operations to accepted evidence, allowing the agent to preserve a valid prefix and repair only an invalidated workflow suffix. The training pipeline specializes the Planner for workflow composition and the Executor for runtime argument grounding.

Earth-Bench-Pro exposes a consistent difficulty gap: Open-World Execution remains the hardest regime across backbones (Figure 2). On Earth-Bench-OW, with a shared GPT-5 backbone, tool set, and skill set, Earth-Agent-Pro reaches 66.13% accuracy under a large language model (LLM) judge, 20.95

percentage points above ReAct, and leads ReAct by 24.44 points on Tools-In-Order. Combining the Planner and Executor adapters raises Qwen3.5-9B LLM-as-Judge accuracy by 11.69 points; planning-only and oracle-plan evaluations separately measure workflow composition and argument grounding.

Our contributions are:

- **Open-world execution task and benchmark.** We formulate open-world EO execution and operationalize it with 248 Open-World Execution questions in Earth-Bench-Pro. Product and Spectrum tasks require runtime observation acquisition and preparation, and all questions require open-ended answers grounded in execution evidence. Together with 496 matched Instruction-Following and Autonomous-Planning questions, the released suite contains over 700 evaluation questions.
- **Open-world EO agent framework.** Earth-Agent-Pro uses expert skill guidance to constrain workflow construction and runtime tool use, while its execution-adaptive dynamic workflow maintains the plan, accepted evidence, and their dependencies. Under a shared backbone, tool set, and skill set, Earth-Agent-Pro outperforms ReAct, AFlow, and OpenEarthAgent on final-answer and workflow-order metrics in controlled comparisons.
- **Dependency-preserving data and role training.** From 248 seed workflows, we construct 2,480 dependency-preserving trajectories and combine them with examples from the public OpenEarthAgent training set [25] to form a balanced mixture for Planner SFT and Executor GRPO. Combining the two adapters raises Qwen3.5-9B LLM-as-Judge accuracy by 11.69 points on Earth-Bench-OW.

II. RELATED WORK

A. Earth Observation Benchmarks and Datasets

Earth observation (EO) datasets provide perception and multimodal evaluation on fixed inputs [32]. RSVQA [27] and VRSBench [28] assess visual question answering; GeoBench-VLM [33] and CHOICE [34] evaluate geospatial multimodal

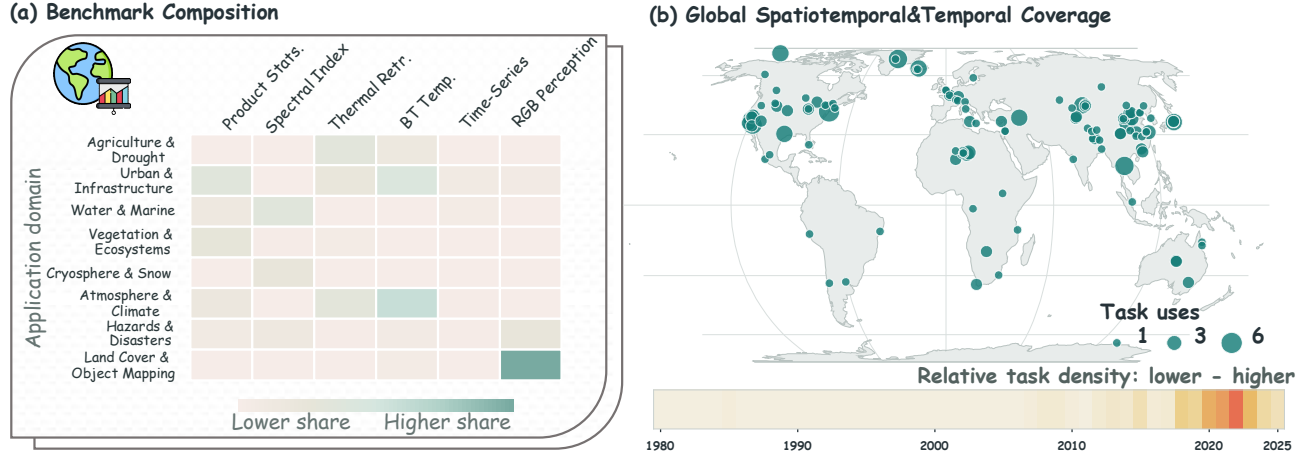


Fig. 3. **Earth-Bench-Pro Overview.** (a) Task composition across application domains and task families. (b) Global spatial distribution and temporal coverage.

capabilities; VLEO-Bench [35], XLRs-Bench [36], URBench [4], DisasterM3 [37], and TerraScope [21] extend evaluation to fine-grained, ultra-high-resolution, urban, disaster, and pixel-grounded reasoning. These benchmarks mainly score final outputs rather than agent interactions, tool-use trajectories, or intermediate artifacts. Complementary EO benchmarks target specific agentic operations: RescueADI [38] evaluates sequential disaster interpretation; GeoLLM-Engine [39] provides analyst-style API and interface interactions; GeoBenchX [29] tests multistep GIS tool calling; AEOS-Bench [40] considers constrained constellation scheduling. Agent-oriented benchmarks formalize these capabilities in their protocols: ThinkGeo [22] assesses tool use, Earth-Bench [31] evaluates trajectory-level EO tasks, and HTAM/GeoPlan-Bench [24] targets hierarchical task abstraction. More recently, EO-Gym [41] and TerraBench [42] introduce interactive heterogeneous data; GeoMMBench [43] and TerraLogic [44] target complex geoscientific reasoning; and GeoAgentBench [45] and DORA [46] assess GIS execution verification and emergency geospatial reasoning. OpenEarth-Bench [30] evaluates open-environment EO pipelines and tool creation, whereas NASA-EO-Bench [47] focuses on data and tool discovery. Yet evaluations often start from curated cases or treat discovery separately, leaving open the joint assessment of data-source selection, acquisition, intermediate evidence, and final natural-language answers.

B. Earth Observation Methods

MLLM-based EO Methods. EO representation and perception methods, including SpectralGPT [48], RingMo-Aerial [49], REST [50], RingMoE [51], CrossEarth [52], and HyperSIGMA [53], have strengthened representation learning and visual interpretation across spectral, aerial, and hyperspectral observations [54]. In parallel, RSGPT [55], SkySenseGPT [56], GeoChat [19], LHRs-Bot [20], EarthGPT [57], EarthMarker [58], EarthGPT-X [59], EarthDial [60], VHM [61], GeoPixel [62], and SkyEyeGPT [63], extend captioning and visual question answering to scene understanding, grounding, multi-sensor dialogue, and pixel-level reasoning [64], [65]. These advances strengthen understanding of provided imagery

or products but generally do not discover external data, coordinate tool use, or maintain evidence through long-horizon EO analysis.

Agent-based EO Methods. General agents establish planning and tool-use paradigms through reasoning-action interleaving in ReAct [66], API-grounded tool use in Gorilla [67] and ToolLLM [68], reusable skills in Voyager [69], and workflow optimization in AFlow [70]. Early EO agents, including RS-ChatGPT [71], RS-Agent [72], and Change-Agent [73], connect LLMs with visual models or expert tools for perception tasks. Subsequent systems organize broader workflows through EO code generation in UnivEarth [23], tool use in ThinkGeo [22], MCP-based tool integration in Earth-Agent [31], knowledge-flow fusion in CangLing-KnowFlow [74], and hierarchical abstraction in HTAM [24]. Recent directions extend to multi-platform reasoning with RingMo-Agent [75], climate science with EarthLink [76], open geospatial environments with OpenEarthAgent [25] and OpenEarth-Agent [30], workflow automation with GeoFlow [77], experience-driven cooperation with GeoEvolver [78], vague-intent handling with RemoteAgent [79], and expert-level reasoning with GeoMMAgent [43]. However, existing systems commonly plan over broad or predefined tool spaces and optimize planning and execution as a unified policy. They rarely use expert task skills to constrain both workflow construction and runtime tool scope, or explicitly maintain accepted evidence and dependencies as execution evolves. Role-specific alignment of workflow composition and node-level argument grounding therefore remains largely underexplored.

III. EARTH-BENCH-PRO BENCHMARK

A. Benchmark Overview

Earth-Bench-Pro builds a 744-question EO agent benchmark from 248 matched scientific task cores. Earth-Bench-Pro comprises three matched sub-benchmarks: Earth-Bench-IF, Earth-Bench-AP, and Earth-Bench-OW. Earth-Bench-IF provides an ordered reference tool sequence and prepared inputs. Earth-Bench-AP hides the sequence while retaining prepared inputs. Earth-Bench-OW requires agents to acquire or discover data

TABLE I

COMPARISON WITH RECENT EARTH-OBSERVATION AGENT BENCHMARKS. VALUES MARKED WITH * ARE NOT EXPLICITLY REPORTED IN THE ORIGINAL PAPERS AND ARE COMPUTED FROM THE CORRESPONDING OFFICIAL OPEN-SOURCE RELEASES.

Benchmark	Tasks	Steps	Avg.	Tools	Multi-source EO Data	Open-ended QA	Executable Trajectory	Real Data Acquisition	Preprocessing Required	Quantitative Answer	Intermediate Observations
Ours	744	5,295	7.1	112	✓	✓	✓	✓	✓	✓	✓
OpenEarth-Bench [30]	596	–	–	–	✓	✗	✓	✓	✓	✓	Part.
OpenEarthAgent [25]	1,169	7,064	6.04	24	✓	✓	✓	Part.	Part.	✓	✓
KnowFlow-Bench [74]	324	–	–	–	✓	✗	✓	Part.	✓	✓	Part.
TerraBench [42]	403	~24,500	~60.8	77	✓	✓	✓	✓	Part.	✓	✓
EO-Gym [41]	1,436	5,323*	3.7*	35	✓	✓	✓	Part.	Part.	Part.	✓
GeoPlan-Bench [24]	996	10,619*	10.7*	104*	✗	✗	✗	✗	✓	✗	✗

at runtime and subsequently generate an open-ended answer grounded in execution evidence.

As illustrated in Figure 3, Earth-Bench-Pro covers eight application domains and six analysis types. The application domains include agriculture and drought [80], urban areas and infrastructure, water and marine systems, vegetation and ecosystems [81], cryosphere and snow, atmosphere and climate, hazards and disasters, and land cover and object mapping. The analysis types include product statistics, spectral indices, thermal retrieval, brightness-temperature analysis, time-series analysis, and RGB perception. The tasks also span regions across all six inhabited continents and requested observation periods from 1980 to 2025, covering a broad range of geographic settings and temporal extents.

Recent EO agent benchmarks cover only parts of the end-to-end workflow. OpenEarth-Bench [30] and KnowFlow-Bench [74] include acquisition or preprocessing but omit open-ended answers. OpenEarthAgent [25], TerraBench [42], and EO-Gym [41] evaluate open-ended answers and tool trajectories but provide partial coverage of acquisition, preprocessing, or quantitative analysis. GeoPlan-Bench [24] isolates workflow planning without executable reference trajectories or runtime acquisition. As shown in Table I, Earth-Bench-Pro is the only benchmark in this comparison that covers all seven dimensions. This design evaluates both final answers and the complete execution processes that produce them.

B. Problem Formulation

Earth-Bench-Pro represents each benchmark sample as $z = (q, c, \tau^*, y^*)$, where q is a scientific query, c encodes the input context or runtime data requirements, and y^* is the reference answer. Let N_z denote the number of steps in the reference trajectory τ^* . The EO environment $\mathcal{E}$ executes these steps. Each step records tool t_i^* , arguments θ_i^* , and the resulting observation o_i^* :

$$\tau^* = ((t_i^*, \theta_i^*, o_i^*))_{i=1}^{N_z}, \quad o_i^* = \mathcal{E}(t_i^*, \theta_i^*). \quad (1)$$

The three evaluation regimes expose different information to the agent. Let $\pi^* = (t_1^*, \dots, t_{N_z}^*)$ denote the reference tool sequence, and let ϕ construct an instruction that states

these tools in order alongside the scientific query. Instruction following uses the resulting instruction $q^{\text{IF}} = \phi(q, \pi^*)$ and prepared inputs c^{prep} . Autonomous planning uses the query without reference tools and the same prepared inputs. Open-world execution provides runtime input requirements c^{req} , leaving the agent to acquire or discover data during execution:

$$\begin{cases} q^{\text{IF}} = \phi(q, \pi^*), \\ \mathcal{I}_{\text{IF}} = (q^{\text{IF}}, c^{\text{prep}}), \\ \mathcal{I}_{\text{AP}} = (q, c^{\text{prep}}), \\ \mathcal{I}_{\text{OW}} = (q, c^{\text{req}}). \end{cases} \quad (2)$$

In the open-world execution, f denotes the tool-selection policy, g denotes the runtime argument generator, and h denotes the evidence-conditioned answer function. The tool and arguments for call i depend on the query, runtime requirements, and preceding accepted execution trace. The environment returns a new observation, and the agent generates its answer from the completed trajectory:

$$\begin{cases} \tau = ((t_i, \theta_i, o_i))_{i=1}^L, \\ t_i = f(q, c^{\text{req}}, \tau_{<i}), \\ \theta_i = g(q, c^{\text{req}}, t_i, \tau_{<i}), \\ o_i = \mathcal{E}(t_i, \theta_i), \\ \hat{y} = h(q, c^{\text{req}}, \tau). \end{cases} \quad (3)$$

Execution therefore directly determines the selected data, intermediate dependencies, and ultimately the final answer.

C. Data Curation

Earth-Bench-Pro treats a complete executable evidence chain as the unit of annotation, linking each scientific request to its answer through tool execution. We assembled an eight-member annotation team with two computer science experts, three remote sensing specialists, and three Earth science specialists. The team jointly reviews tool executability, remote sensing data requirements, scientific interpretation, and answer correctness, bringing technical constraints and domain judgment into the same annotation process.

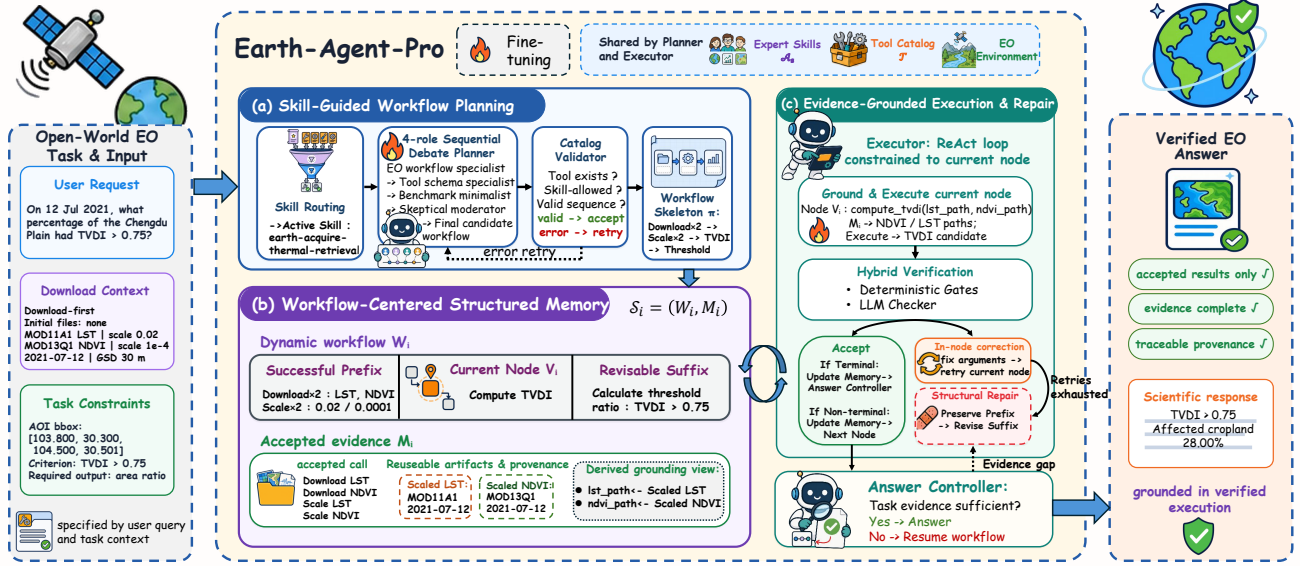


Fig. 4. **Earth-Agent-Pro Framework.** (a) Skill-guided workflow planning uses a shared expert specification to constrain both workflow construction and runtime argument grounding. (b) Workflow-centered structured memory couples the evolving workflow with provenance-aware accepted evidence, preventing failed attempts from contaminating downstream execution. (c) Evidence-grounded execution and repair performs node-local correction and, upon persistent failure or an evidence gap, preserves the successful prefix while revising only the affected workflow suffix. Together, these mechanisms enable traceable and bounded adaptation without discarding valid execution progress.

Experts transform 248 scientific task cores into open-world workflows for which execution determines the inputs, intermediate dependencies, and answers. These tasks retain the scientific objectives of Earth-Bench [31]. Experts re-execute the original solutions in a shared EO execution environment and correct inconsistencies among the question, tool selection, call arguments, intermediate dependencies, and answer. For 188 Product and Spectrum tasks, experts replace fixed local inputs with online acquisition and required preprocessing, authoring 323 task-level acquisition specifications. Each specification encodes the required source, product, bands, spatial extent, observation period, resolution, sampling interval, and scaling information. The remaining 60 RGB tasks retain user-provided visual inputs, while concrete files still require runtime discovery. After reconstruction, file paths and intermediate artifacts become execution outcomes, and downstream calls must bind the outputs returned by upstream tools.

Each reconstructed task pairs an open-ended query with reference evidence regenerated through execution. Experts remove all answer choices and rewrite each question as a self-contained request specifying the scientific objective, study region, observation period, and task constraints. They rerun the workflow in the same environment, record every call and returned observation, and recompute the answer from the outputs. Each sample retains the representation introduced in Section III:

$$z = (q, c, \tau^*, y^*), \quad \tau^* = ((t_i^*, \theta_i^*, o_i^*))_{i=1}^{N_z}. \quad (4)$$

Here, context c records runtime input requirements, and each reference step records tool t_i^* , arguments θ_i^* , and observation o_i^* . This representation preserves artifact provenance, data dependencies, and answer evidence along the complete execution path.

Quality control checks three relations along the evidence chain: queries against input requirements, data dependencies across tool calls, and final answers against execution evidence. Experts retain and inspect intermediate rasters, files, numerical outputs, detections, masks, and statistics. They also verify answer values, units, precision, temporal references, and trend descriptions. The 248 verified Open-World Execution workflows contain 1,765 tool calls, average 7.1 calls per task, and use 84 distinct tools. We derive matched Instruction-Following and Autonomous-Planning variants from each verified task core, yielding 744 questions and 5,295 reference tool calls. Every answer retains a traceable execution basis, and every intermediate result records its upstream source.

IV. EARTH-AGENT-PRO: AN EXECUTION-ADAPTIVE AGENT FRAMEWORK

A. Framework Overview

Earth-Agent-Pro is a Plan-and-Execute-style agent that separates scientific workflow planning from runtime operation grounding. Before concrete files and intermediate results become available, the Planner constructs an ordered workflow skeleton. As data and intermediate artifacts emerge, the Executor grounds each node, invokes its tool, and updates the state. The dynamic workflow and accepted execution memory form the workflow-centered structured memory shared by both roles throughout the entire execution process. The Planner and workflow controller operationalize the abstract tool-selection policy f in Eq. (3), the argument generator G_{exec} implements g , and the Answer Controller implements h .

Figure 4 presents the complete control process. Let R_{skill} and R_{tool} denote skill and tool routing, let $\mathcal{T}$ denote the global tool catalog, and let $\mathcal{A}_s$ denote the tools allowed by skill s . The two routing stages reduce the catalog to the candidate set

$\mathcal{P}_s$, after which the Planner generates the tool sequence π^{plan} through sequential debate. The initialization operator converts this sequence into dynamic workflow W_0 and creates the initial state S_0 with empty memory M_0 :

$$\begin{cases} s = R_{\text{skill}}(q, c), \\ \mathcal{P}_s = R_{\text{tool}}(q, c, s, \mathcal{T} \cap \mathcal{A}_s), \\ \pi^{\text{plan}} = \text{Plan}(q, c, s, \mathcal{P}_s), \\ S_0 = (W_0, M_0) = (\text{Init}(\pi^{\text{plan}}), \emptyset). \end{cases} \quad (5)$$

The Executor advances node by node over this structured state and exposes only verified results to subsequent operations. When a node fails or the final evidence remains incomplete, the Planner proposes a local replacement suffix and the Executor installs it while retaining the successful prefix.

B. Skill-Guided Workflow Planning

Expert skills encode EO procedure knowledge as task-level specifications shared by the Planner and Executor. We author six skills covering spectral analysis, product-based computation, and RGB perception. Each skill describes its tool scope, operation order, argument constraints, and repetition rules across acquisition, preprocessing, domain computation, and aggregation. As illustrated in Figure 5, skill guidance removes irrelevant tools before workflow generation and continues to constrain node realization during execution.

The Skill Router and Tool Router implement the two constraints in Eq. (5). The full catalog $\mathcal{T}$ contains 112 callable tools and their schemas. The Skill Router selects a skill from the query and input context, then excludes inapplicable tools through $\mathcal{A}_s$. The Tool Router retrieves $\mathcal{P}_s$ for the task objective, leaving the Planner to compose a workflow over legal and task-relevant tools for subsequent execution.

The candidate tools must form a scientifically valid and interface-compatible operation sequence. The Planner uses a four-role sequential debate that examines the EO procedure, tool schemas, sequence parsimony, and final executability in order. Each role revises the candidate sequence using the preceding discussion. A catalog validator then checks tool existence and active-skill membership. It returns validation errors with the candidate sequence for another planning attempt until the Planner obtains π^{plan} .

The selected skill remains active after initial planning. For the current operation, the Executor combines the skill instructions with the tool schema, query constraints, and accepted upstream results to generate arguments. When deterministic rules cannot resolve a tool result, the language-model checker uses the same skill to assess whether the result follows the task procedure. The Planner and Executor therefore share one task-level specification for operation selection and runtime argument grounding.

C. Workflow-Centered Structured Memory

Earth-Agent-Pro represents runtime step k using the structured state $S_k = (W_k, M_k)$. The dynamic workflow W_k contains the active node sequence $\mathcal{V}_k$ and the revision archive $\mathcal{R}_k$. The execution memory M_k is a chronologically ordered

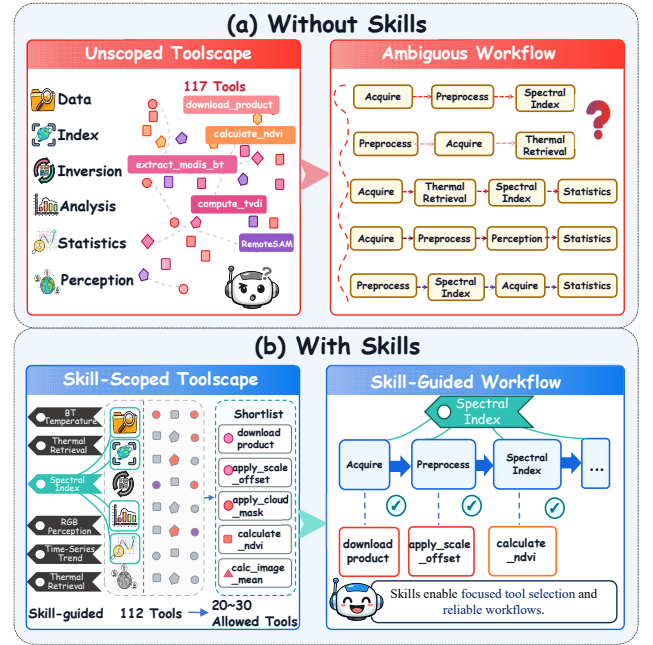


Fig. 5. **With vs. Without Skills.** The top shows an unscoped tool space and ambiguous workflows without skills, while the bottom shows focused tool selection and a coherent workflow enabled by skill guidance.

collection of evidence records. Each record e_u stores invoked tool $\bar{t}_u$, arguments θ_u , observation o_u , and provenance ν_u :

$$\begin{cases} S_k = (W_k, M_k), \\ W_k = (\mathcal{V}_k, \mathcal{R}_k), \\ \mathcal{V}_k = (v_1^k, \dots, v_{n_k}^k), \\ M_k = (e_1, \dots, e_{m_k}), \end{cases} \quad e_u = (\bar{t}_u, \theta_u, o_u, \nu_u). \quad (6)$$

Each node records its planned tool, invoked tool, execution status, grounded arguments, result summary, failure diagnosis, and revision identifier. Execution memory accumulates only verified calls. Provenance for file artifacts also records variables, time, region, bands, and batches, allowing the Executor to resolve concrete files, numerical series, and cross-input correspondence for the later nodes.

The structured state limits the information exposed to subsequent operations at each step. Argument generation uses the query, current node and schema, active skill, and previously accepted call history. Rejected attempts provide retry feedback for the current node and leave the accepted history unchanged. The Executor therefore avoids repeatedly recovering the valid evidence, failed attempts, and pending operations from an expanding free-form interaction transcript.

D. Evidence-Grounded Execution and Workflow Repair

The Executor advances the workflow through a ReAct loop constrained to the current node. For node v_i^k in state S_k , let t_i^k denote its planned tool, let a index in-node attempts, and let $\bar{t}_{i,a}^k$ denote the invoked tool. The first attempt sets $\bar{t}_{i,1}^k = t_i^k$. $\sigma(t)$ returns the schema of tool t . Argument generator G_{exec} combines this schema, the current node, and accepted memory to generate arguments. Execution environment $\mathcal{E}$ returns an

observation, after which hybrid verification gate V_{hyb} produces decision $\delta_{i,a}^k$ and feedback $b_{i,a}^k$:

$$\begin{cases} \hat{\theta}_{i,a}^k = G_{\text{exec}}(q, c, s, \sigma(\tilde{t}_{i,a}^k, v_i^k, W_k, M_k), \\ \hat{\delta}_{i,a}^k = \mathcal{E}(\tilde{t}_{i,a}^k, \hat{\theta}_{i,a}^k), \\ (\delta_{i,a}^k, b_{i,a}^k) = V_{\text{hyb}}(q, s, W_k, M_k, \tilde{t}_{i,a}^k, \hat{\theta}_{i,a}^k, \hat{\delta}_{i,a}^k), \\ \delta_{i,a}^k \in \{\text{accept, retry, replace}\}. \end{cases} \quad (7)$$

The hybrid gate first applies deterministic checks for structural errors, error markers, and batch coverage. The language-model checker evaluates cases that these rules cannot resolve using the query, grounded arguments, active skill, and remaining workflow. Retry updates the arguments and feedback for the current tool. Replace selects another tool from $\mathcal{P}_s$ and updates the feedback. Both actions remain local to the current attempt and leave persistent state $\mathcal{S}_k$ unchanged. Accept instead commits the node result to the dynamic workflow and appends its evidence record to execution memory. Let $\nu_{i,a}^k$ denote call provenance, let U_{acc} denote node commitment, and let append add an item to an ordered collection. After all in-node attempts fail, the Executor writes the final diagnosis to the failed node and then triggers localized workflow repair:

$$\begin{aligned} e_{i,a}^k &= (\tilde{t}_{i,a}^k, \hat{\theta}_{i,a}^k, \hat{\delta}_{i,a}^k, \nu_{i,a}^k), \\ \mathcal{S}_{k+1} &= \begin{cases} (U_{\text{acc}}(W_k, i, e_{i,a}^k), \\ \text{append}(M_k, e_{i,a}^k)), & \text{if } \delta_{i,a}^k = \text{accept}, \\ \mathcal{S}_k, & \text{if } \delta_{i,a}^k \in \{\text{retry, replace}\}. \end{cases} \end{aligned} \quad (8)$$

The Planner generates candidate suffix $\tilde{\mathcal{V}}_i$ from failed node i , where d_i^k denotes the final failure diagnosis stored by that node. The Executor validates candidate operations against the active skill and tool catalog. It rejects candidates that repeat an earlier repair, show no execution progress, have abnormal length, or contain excessive operation repetition. After a candidate passes these checks, the Executor uses $\oplus$ to concatenate the successful prefix and new suffix in execution order, then records the old suffix with its diagnosis in $\mathcal{R}_{k+1}$. The repair leaves M_k unchanged and re-executes only the failed position and subsequent operations:

$$\begin{cases} \tilde{\mathcal{V}}_i = \text{Plan}_{\text{repair}}(q, c, s, W_k, M_k, i, d_i^k), \\ \mathcal{V}_{k+1} = \mathcal{V}_k[1:i-1] \oplus \tilde{\mathcal{V}}_i, \\ \mathcal{R}_{k+1} = \text{append}(\mathcal{R}_k, (\mathcal{V}_k[i:], \tilde{\mathcal{V}}_i, d_i^k)), \\ W_{k+1} = (\mathcal{V}_{k+1}, \mathcal{R}_{k+1}), \\ M_{k+1} = M_k. \end{cases} \quad (9)$$

The Answer Controller extends the same repair mechanism to workflows whose tools succeed but whose evidence remains incomplete. It checks required computations, spatiotemporal coverage, aggregation, units, numerical scale, and non-finite results. With sufficient evidence, the controller generates the final answer only from M_k . Incomplete evidence produces an actionable diagnosis, after which the Executor reopens the specified node or defaults to the final node when the diagnosis identifies none. Earth-Agent-Pro therefore uses a node as the boundary for runtime correction and a workflow suffix as the boundary for structural revision, respectively.

V. ROLE-SPECIALIZED TRAINING

A. Training Overview

The Planner and Executor expose different prediction interfaces. The Planner composes an ordered tool sequence from expert skill guidance and candidate tool schemas. The Executor grounds a prescribed node from accepted runtime evidence. A monolithic trajectory objective would mix workflow errors with argument errors and spread credit across long tool chains. Role-specialized training isolates these errors at their respective prediction interfaces.

The training pipeline expands instances through dependency-aware trajectory instantiation and then extracts role-specific supervision from complete trajectories. Let $\mathcal{D}_P$ and $\mathcal{D}_E$ denote the Planner and Executor records. Parameters ψ_P and ψ_E are the trainable low-rank adaptation (LoRA) parameters of the two role-specific adapters. Planner context x^P contains the query, inputs, skill guidance, and candidate tool schemas. Executor context x_i^E contains the prescribed node and accepted reference-trajectory prefix. For a Planner record, ω^P tokenizes reference tool sequence π^* . For Executor sample j at node i , $\omega_{i,j}^E$ is the response token sequence.

Supervised fine-tuning (SFT) optimizes workflow composition, while on-policy group relative policy optimization (GRPO) [82] optimizes runtime argument grounding. Let $\hat{A}_{i,j}$ denote group-relative advantage, and let $\rho_{i,j,\ell}$ and $\bar{\rho}_{i,j,\ell}$ denote the raw and clipped token-level importance ratios. The frozen reference distribution is p_{ref} . Coefficients β and η weight the Kullback–Leibler (KL) regularizer and entropy bonus, respectively. The total objective combines the two role losses, and we optimize a separate adapter for each role:

$$\begin{cases} \mathcal{L}_{\text{plan}}(\psi_P) = -\mathbb{E}_{\mathcal{D}_P} \sum_{\ell} \log p_{\psi_P}(\omega_{\ell}^P | x^P, \omega_{<\ell}^P), \\ \mathcal{L}_{\text{exec}}(\psi_E) = -\mathbb{E}_{i,j,\ell} \left[\min(\rho_{i,j,\ell} \hat{A}_{i,j}, \bar{\rho}_{i,j,\ell} \hat{A}_{i,j}) \right] \\ \quad + \beta \hat{D}_{\text{KL}}(p_{\psi_E} \| p_{\text{ref}}) - \eta \mathcal{H}(p_{\psi_E}), \\ \mathcal{L}_{\text{train}}(\psi_P, \psi_E) = \mathcal{L}_{\text{plan}}(\psi_P) + \mathcal{L}_{\text{exec}}(\psi_E), \\ (\psi_P^*, \psi_E^*) \in \arg \min_{\psi_P, \psi_E} \mathcal{L}_{\text{train}}. \end{cases} \quad (10)$$

B. Dependency-Preserving Training Data Construction

Training data follow the executable sample representation of Earth-Bench-Pro. A seed sample is $z = (q, c, \tau^*, y^*)$ with $\tau^* = ((t_i^*, \theta_i^*, o_i^*))_{i=1}^{N_z}$. We decompose it into an immutable execution skeleton $\kappa(z)$ and a mutable instance state $\mathbf{u}(z)$. The skeleton retains the reference tool sequence π^* , the corresponding tool schemas, and dependency structure $\mathcal{G}_z$, a mathematical abstraction of the dependencies encoded by slot groups and cross-node references. The instance state contains locations, spatial extents, temporal ranges, sample identifiers, file references, and replaceable intermediate observations.

For each seed sample, a language model represents the mutable state as typed slots, groups repeated references and dependent fields, and generates a sample-specific instantiation program F_z . Given root-slot values $\tilde{\mathbf{u}}_{\text{root}}$, the program propagates replacements along these dependencies to produce $\tilde{z}$ while preserving $\kappa(\tilde{z}) = \kappa(z)$. It therefore instantiates coupled

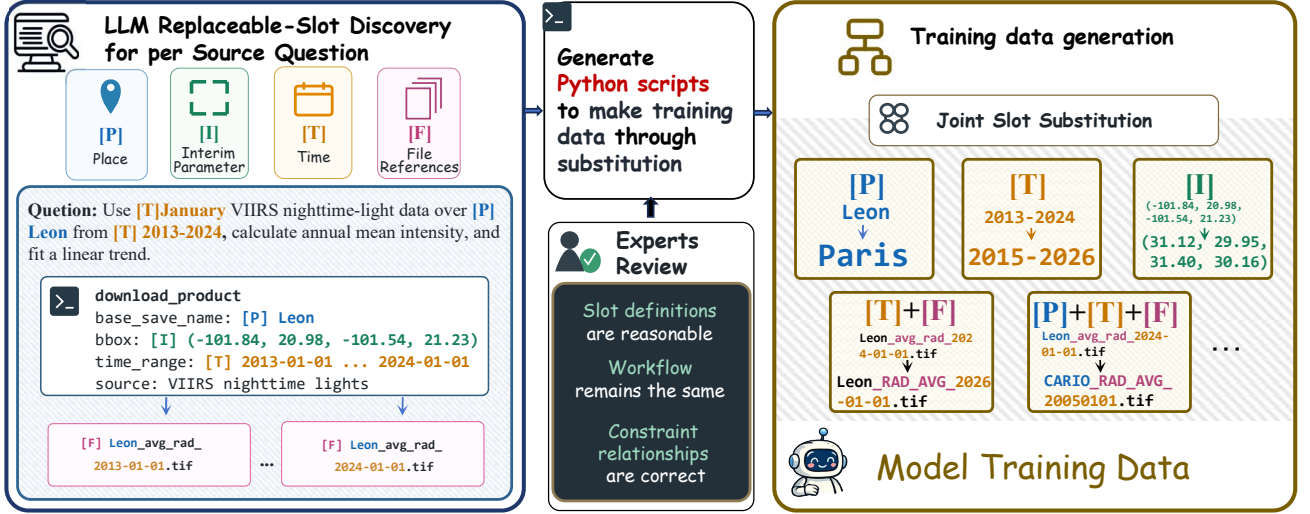


Fig. 6. **Synthetic training data generation.** An LLM identifies replaceable slots for each seed sample and writes a sample-specific Python program. Human experts verify that the program preserves workflow structure and dependency constraints before it generates consistent training instances.

fields such as a place and its bounding box together and propagates changed node outputs to all downstream arguments and observations. Human reviewers inspect the slot specification and program before batch synthesis. Figure 6 shows the process from a seed sample through a reviewed instantiation program to dependency-consistent training instances. Let R denote the number of mutable slots. Operator Compose reconstructs a sample from its execution skeleton and instance state. We formalize this construction as follows:

$$\begin{cases} \kappa(z) = (\pi^*, \{\sigma(t_i^*)\}_{i=1}^{N_z}, \mathcal{G}_z), \\ \mathbf{u}(z) = (u_1, \dots, u_R), \\ z = \text{Compose}(\kappa(z), \mathbf{u}(z)), \\ \tilde{z} = F_z(\kappa(z), \tilde{\mathbf{u}}_{\text{root}}), \\ \kappa(\tilde{z}) = \kappa(z). \end{cases} \quad (11)$$

The reviewed programs fix the sensor or data product, selected bands, ordered tools, argument schemas, processing operations, and dependency structure. They then instantiate samples under a fixed random seed and apply quality filtering. This process yields 2,480 instantiated trajectories derived from 248 seed workflows across RGB imagery, spectral observations, and remote sensing products. We add examples from the OpenEarthAgent training set [25] to cover GIS analysis, object-centric visual perception, and geometric reasoning, then carefully balance the two sources to form the final trajectory set $\mathcal{Z}_{\text{train}}$. The Planner extractor creates one sequence-level record per trajectory with target π^* . The Executor extractor creates one argument record per node with target θ_i^* :

$$\begin{cases} \mathcal{D}_P = \{(x^P(z), \pi^*(z)) \mid z \in \mathcal{Z}_{\text{train}}\}, \\ \mathcal{D}_E = \{(x_i^E(z), \theta_i^*(z)) \mid z \in \mathcal{Z}_{\text{train}}, 1 \leq i \leq N_z\}. \end{cases} \quad (12)$$

C. Planner SFT for Workflow Composition

Planner SFT maps domain-constrained context x^P to reference workflow π^* . In Eq. (10), $\mathcal{L}_{\text{plan}}$ computes autoregressive negative log-likelihood over the assistant response serializing

π^* . This response contains no tool arguments or observations, so the objective matches the Planner’s inference interface.

Each Planner example uses a single-turn conversation format. The system and user messages form x^P and provide the scientific query, input context, skill guidance, and candidate tools. The assistant target serializes π^* as a JSON object containing only the `tool_sequence` field. We mask prompt tokens, so $\mathcal{L}_{\text{plan}}$ supervises only the assistant response and focuses adaptation on tool selection and ordering.

Parameter-efficient adaptation focuses the Planner’s learning signal on its role-specific workflow decisions. We freeze the pretrained backbone throughout the entire optimization process and update only a Planner-specific low-rank adaptation (LoRA) module [83] to learn EO tool-selection and ordering patterns. At inference, the sequential debate and catalog validation described in Section IV enforce procedure, schema, and active-skill constraints beyond the learned workflow prior.

D. Executor GRPO for Runtime Argument Grounding

Executor training treats the current workflow node as a local decision interface. Given node context x_i^E and the schema of prescribed tool t_i^* , policy p_{ψ_E} generates argument dictionary $\hat{\theta}_i$ at inference and sampled dictionaries $\theta_{i,j}$ during GRPO. The reward evaluates only argument structure and values because the workflow already fixes the tool identity. This node-level interface isolates feedback for argument grounding from workflow decisions at other nodes.

The Executor reward compares each parsed response directly with the reference argument dictionary θ_i^* and requires no learned reward model. Let `parse` denote the JSON parser, let $\perp$ denote parse failure, let $\mathbb{J}$ denote the set of valid JSON values, and let $\mathbb{D}_+ \subset \mathbb{J}$ denote the set of nonempty dictionaries. For candidate j at node i , $\omega_{i,j}^E$ denotes the raw token sequence, $\xi_{i,j} = \text{parse}(\omega_{i,j}^E)$ denotes its parsed value, and we set $\hat{\theta}_{i,j} = \xi_{i,j}$ when $\xi_{i,j} \in \mathbb{D}_+$. For a nonempty dictionary, K^* and $\hat{K}$ denote the reference and predicted key sets, $K_c = K^* \cap \hat{K}$ and $K_d = K^* \triangle \hat{K}$ denote the

common and discrepant keys, and $m = |K^*| > 0$. Value-agreement term A_{val} counts type-aware matches over common keys under relation $\equiv$, which recursively compares nested values and their types with a tolerance of 10^{-6} for floating-point values. The following equation combines key coverage, a symmetric-difference penalty, and value agreement in S_{arg} , then defines complete reward $r_{i,j}$. A missing key incurs a larger penalty than a superfluous key because it reduces coverage and increases the symmetric difference. Runtime verification separately determines execution validity:

$$\begin{aligned} A_{\text{val}} &= \sum_{k \in K_c} \mathbb{I}[\hat{\theta}_{i,j,k} \equiv \theta_{i,k}^*], \\ S_{\text{arg}}(\hat{\theta}_{i,j}, \theta_i^*) &= \max \left\{ -1, \frac{|K_c| - |K_d| + A_{\text{val}}}{m} \right\}, \\ r_{i,j} &= \begin{cases} -2, & \xi_{i,j} = \perp, \\ -1, & \xi_{i,j} \in \mathbb{J} \setminus \mathbb{D}_+, \\ S_{\text{arg}}(\hat{\theta}_{i,j}, \theta_i^*), & \xi_{i,j} \in \mathbb{D}_+. \end{cases} \end{aligned} \quad (13)$$

For each node input, we sample $G = 4$ candidate argument dictionaries from the current policy and standardize their rewards within the group. Let $r_{i,j}$ denote the reward for candidate j , let p_{old} denote the policy that generated it, and let ϵ_c denote the GRPO clipping radius. Relative advantage $\hat{A}_{i,j}$ supplies a quality signal under the same node context without a separate critic. In Eq. (10), the KL regularizer limits deviation from the reference model, and the entropy bonus discourages premature collapse. During training, we freeze the pretrained backbone and update only the Executor-specific LoRA adapter. The trained policy grounds arguments for the current node, while explicit runtime mechanisms control environment interaction, observation verification, retries, and workflow repair. The group-relative advantage, token-level importance ratio, and its clipped form define the GRPO surrogate term:

$$\begin{cases} \hat{A}_{i,j} = \frac{r_{i,j} - \text{mean}_{j'}(r_{i,j'})}{\text{std}_{j'}(r_{i,j'}) + 10^{-6}}, \\ \rho_{i,j,\ell} = \frac{p_{\psi_E}(\omega_{i,j,\ell}^E \mid x_i^E, \omega_{i,j,<\ell}^E)}{p_{\text{old}}(\omega_{i,j,\ell}^E \mid x_i^E, \omega_{i,j,<\ell}^E)}, \\ \bar{\rho}_{i,j,\ell} = \text{clip}_{[1-\epsilon_c, 1+\epsilon_c]}(\rho_{i,j,\ell}). \end{cases} \quad (14)$$

VI. EXPERIMENTS

We evaluate Earth-Agent-Pro from seven perspectives. We first examine its sensitivity to LLM backbones and compare it with remote-sensing MLLMs. We then assess cross-benchmark transfer on the OpenEarthAgent benchmark [25] and compare Earth-Agent-Pro with general-purpose agents. Next, we isolate the effect of the agent framework through controlled comparisons on Earth-Bench-Pro. Finally, we ablate domain skills and disentangle the contributions of Planner SFT and Executor GRPO [82] through adapter composition, planning-only evaluation, and oracle-plan execution. Together, these experiments evaluate the effectiveness and transferability of Earth-Agent-Pro and identify the mechanisms.

A. Experimental Setup

a) Benchmarks and Comparison Groups: Our main evaluation is conducted on Earth-Bench-Pro (Section III), which

converts Earth-Bench into open-ended, end-to-end Earth-observation tasks that require data acquisition, preprocessing, analysis, and evidence-grounded answer generation. We additionally evaluate transfer to the OpenEarthAgent benchmark [25], compare against remote-sensing MLLMs on the categories used by TerraScope [21], and compare with general-purpose agents across Spectrum, Products, and RGB tasks.

b) Baselines and Models: We first evaluate Earth-Agent-Pro with three closed-weight LLM backbones, GPT-5, Gemini-3.5-flash, and Claude-4.6-sonnet, and open-weight backbones from the GLM, Llama, InternLM, GPT-oss, Qwen, Gemma, and Mistral families. We also include fine-tuned Qwen3.5-9B and Qwen3.5-4B variants. We conduct this training and all open-weight model evaluations on eight NVIDIA H200 graphics processing units (GPUs), use vLLM for model serving, and query the three closed-weight models through their official APIs. The remote-sensing comparison includes GeoChat [19], TeoChat [84], EarthDial [60], LHRS-Bot [20], EarthMind [85], and TerraScope [21]. For framework-level evaluation on the OpenEarthAgent benchmark [25], we compare ReAct [66], AFlow [70], OpenEarthAgent [25], and Earth-Agent-Pro under a shared GPT-5 backbone. We next compare against the general-purpose agents Codex and Manus. Finally, we repeat the framework comparison on Earth-Bench-Pro using the same GPT-5 backbone to more clearly isolate the specific effect of the agent framework itself.

c) Metrics: We adopt benchmark-specific evaluation protocols. On Earth-Bench-Pro, we mainly follow the dual-level evaluation protocol of Earth-Agent [31]. Final-answer quality is evaluated using LLM-as-Judge accuracy, ROUGE-L [86], and embedding-based semantic matching (EM), which respectively measure judged correctness, lexical overlap, and semantic similarity to the reference answer. Execution trajectories are evaluated using Efficiency, Tools-Any-Order (TAO), Tools-In-Order (TIO), Tool-Exact-Match (TEM), and Parameter Accuracy. These trajectory metrics collectively characterize tool coverage, tool ordering, strict agreement with the reference workflow, and the correctness of tool arguments.

For the OpenEarthAgent benchmark [25], we retain its official evaluation protocol, including category-wise tool-selection F1, tool-order fidelity, and task-level accuracy. For the final-result comparisons in Tables III and V, we report accuracy only. A non-numerical prediction is counted as correct when it matches the reference answer, while a numerical prediction is considered correct when its relative error with respect to the ground truth is below 15%. All reported accuracy values are computed as the percentage of correct predictions over the evaluated samples for each corresponding comparison.

B. Main Results on Earth-Bench-Pro

a) Backbone and Fine-Tuning Results: We first examine how the policy backbone affects Earth-Agent-Pro on the open-ended Earth-Bench-Pro tasks. We take the off-the-shelf Qwen3.5-4B and Qwen3.5-9B models as our baselines, enabling direct comparison with our fine-tuned variants at the same model scales. Table II reports final answer quality, trajectory length, and step-level agreement with expert workflows under the same unified evaluation protocol.

TABLE II

PERFORMANCE OF DIFFERENT LLM BACKBONES WITH EARTH-AGENT-PRO ON EARTH-BENCH-PRO. ALL QUALITY METRICS ARE PERCENTAGES. EFFICIENCY IS A RATIO FOR WHICH VALUES CLOSER TO 1 ARE BETTER. AMONG OPEN-SOURCE MODELS, BEST RESULTS ARE BOLDED AND RUNNER-UPS ARE UNDERLINED.

Model	LLM-as-Judge	ROUGE-L	EM	Efficiency	TAO	TIO	TEM	Param.
GPT-5	66.13	21.68	13.31	1.0911	87.61	71.97	60.14	27.28
Gemini-3.5-flash	63.31	25.43	20.56	1.0752	89.28	73.90	64.49	32.19
Claude-4.6-sonnet	56.85	29.09	29.03	1.1911	90.17	76.59	63.52	28.02
GLM-4.5-Air	48.39	20.45	<u>20.16</u>	1.0063	85.46	72.76	62.30	27.01
Llama-3.1-8B-Instruct	13.71	5.15	2.02	0.9811	75.34	60.81	44.95	22.04
Llama-3.2-3B-Instruct	4.44	2.25	2.42	0.6873	51.09	44.26	36.62	13.96
Llama-3.3-70B-Instruct	33.87	8.69	6.45	1.0526	85.17	73.51	56.69	27.39
Llama-4-Scout	32.66	9.91	7.66	0.9335	81.81	70.47	57.57	25.88
InternLM3-8B	19.35	6.86	5.24	0.7617	69.33	58.49	48.85	20.73
Intern-S1	40.73	15.49	13.71	<u>1.0022</u>	85.44	73.62	60.24	<u>31.26</u>
GPT-oss-20B	38.71	14.10	10.48	1.1510	85.94	71.08	59.59	28.62
GPT-oss-120B	48.39	13.54	8.87	1.0415	86.16	71.62	61.61	29.84
Qwen3.5-27B	<u>49.60</u>	19.12	15.32	1.0017	87.67	75.27	68.71	29.86
Gemma-4 (E4B-it)	31.45	12.49	9.68	0.7841	69.33	61.36	55.13	23.26
Gemma-4 (26B-A4B-it)	47.58	18.49	14.52	1.0766	84.19	72.62	61.80	31.23
MistralAI-v0.3-7B	14.11	6.32	4.44	0.5876	54.31	46.18	42.18	15.61
Qwen3.5-4B	32.26	16.61	14.52	0.9447	79.21	66.87	57.99	21.58
Qwen3.5-9B	38.31	16.33	15.73	1.0129	85.59	74.57	67.29	25.05
Ours (4B, fine-tuned)	43.15	22.42	21.77	1.1117	<u>87.93</u>	<u>77.52</u>	<u>68.96</u>	28.98
Ours (9B, fine-tuned)	50.00	<u>20.65</u>	19.76	1.1571	91.28	81.11	71.01	32.77

Observation 1: answer and trajectory metrics reveal different model strengths. GPT-5 obtains the highest LLM-as-Judge score (66.13%), whereas the fine-tuned 4B model leads on ROUGE-L and EM. In contrast, the fine-tuned 9B model achieves the strongest TAO, TIO, TEM, and parameter accuracy, showing that final-answer quality and workflow fidelity do not necessarily improve together.

Observation 2: role-specialized training consistently strengthens small backbones. Relative to Qwen3.5-9B, the fine-tuned 9B variant improves LLM-as-Judge accuracy from 38.31% to 50.00% and raises parameter accuracy by 7.72 points. The 4B model exhibits similarly broad gains, including increases of 10.89 points in Judge accuracy and 10.97 points in TEM. These results indicate that the training procedure improves both answer generation and structured tool execution.

C. Comparisons with Existing Methods and Agents

a) *Remote-Sensing MLLM Comparison:* Table III reports the performance of Earth-Agent-Pro and representative remote-sensing MLLMs on five TerraScope task categories, including boundary detection, comparative ranking, coverage estimation, area measurement, and distance measurement. The evaluation uses open-ended questions instead of multiple-choice options, requiring models to directly produce semantic or numerical answers without candidate guidance.

Observation 1. Remote-sensing MLLMs remain weak on quantitative reasoning. Across boundary detection and comparative ranking, existing remote-sensing MLLMs achieve

moderate accuracy, which suggests that these models can often recognize coarse land-cover semantics and compare visible regions. However, their performance drops sharply on coverage estimation, area measurement, and distance measurement, indicating that remote-sensing MLLMs still lack reliable metric grounding when exact numerical answers are required.

Observation 2. Earth-Agent-Pro achieves the highest overall accuracy. Earth-Agent-Pro obtains the best average accuracy among compared methods, reaching 50.58% with Qwen3.5-9B and 50.41% with GPT-5. Compared with the strongest baseline, TerraScope (42.60%), these results correspond to improvements of 7.98 and 7.81 percentage points, respectively. The advantage is most pronounced in area measurement, where both Earth-Agent-Pro variants achieve 31.58%, compared with at most 4.65% among the remote-sensing MLLMs. Earth-Agent-Pro maintains strong semantic performance, achieving 66.43% on boundary detection and up to 71.35% on comparative ranking. These results demonstrate its effectiveness across both semantic understanding and quantitative reasoning tasks in open-ended evaluation settings.

b) Cross-Benchmark Comparison on OpenEarthAgent:

We further evaluate the cross-benchmark transferability of Earth-Agent-Pro on the OpenEarthAgent benchmark [25]. To isolate the effect of the agent framework, we first compare ReAct, AFlow, OpenEarthAgent, and Earth-Agent-Pro using the same GPT-5 backbone and the official OpenEarthAgent benchmark [25] evaluation protocol. We additionally evaluate the role-specialized 9B and 4B variants to examine whether

TABLE III

COMPARISON WITH GENERAL RS MLLMs ACROSS THE TERRASCOPE TASK CATEGORIES. A NUMERICAL PREDICTION IS COUNTED AS CORRECT IF ITS RELATIVE ERROR FROM THE GROUND-TRUTH VALUE IS LESS THAN 15%. THE AVG. COLUMN REPORTS SAMPLE-WEIGHTED ACCURACY ACROSS THE FIVE TASK CATEGORIES. BEST RESULTS ARE IN BOLD, AND SECOND-BEST ARE UNDERLINED.

Model	Boundary Detection (855)	Comparative Ranking (855)	Coverage Estimation (855)	Area Measurement (855)	Distance Measurement (129)	Avg.
GeoChat [19]	57.54	58.83	12.16	0.12	<u>0.78</u>	31.02
TeoChat [84]	56.14	65.85	7.25	0.12	0.00	31.16
EarthDial [60]	55.09	52.98	7.02	0.00	0.00	27.73
LHRS-Bot [20]	55.20	56.73	8.07	1.55	0.00	28.97
EarthMind [85]	54.85	59.77	12.28	<u>4.65</u>	0.00	30.74
TerraScope [21]	<u>66.20</u>	69.94	40.70	0.00	0.00	42.60
Earth-Agent-Pro (GPT-5)	66.43	<u>70.64</u>	<u>40.35</u>	31.58	1.55	<u>50.41</u>
Earth-Agent-Pro (Qwen3.5-9B)	66.43	71.35	<u>40.35</u>	31.58	1.55	50.58

TABLE IV

FRAMEWORK COMPARISON ON THE OPENEARTHAGENT BENCHMARK. WE REPORT TOOL-SELECTION F1 SCORES FOR PERCEPTION (PER.), OPERATION (OP.), LOGIC, AND GIS; TOOL-ORDER FIDELITY UNDER ANYORDER, SAMEORDER, AND UNIQUE; AND TASK-LEVEL ACCURACY FOR NONGENERATIVE ANSWERING (ANS.) AND IMAGE GENERATION (GEN.). BEST RESULTS ARE IN BOLD, AND SECOND-BEST RESULTS ARE UNDERLINED.

Framework	Tool Selection F1				Tool-Order Fidelity			Task Accuracy	
	Per.	Op.	Logic	GIS	AnyOr.	SameOr.	Uni.	Ans.	Gen.
ReAct (GPT-5)	40.87	<u>66.90</u>	17.18	88.41	46.71	46.71	48.59	39.45	80.69
AFlow (GPT-5)	32.54	67.69	36.30	86.75	51.11	50.56	54.17	45.38	65.12
OpenEarthAgent (GPT-5)	16.88	47.00	6.26	91.50	46.96	46.79	47.81	43.88	46.21
OpenEarthAgent-4B	57.49	49.12	44.80	94.51	61.50	60.65	67.32	43.25	71.55
Earth-Agent-Pro (GPT-5)	62.30	55.68	35.15	<u>95.59</u>	64.23	64.80	70.97	56.54	<u>74.01</u>
Ours (4B, fine-tuned)	56.16	63.23	<u>42.92</u>	95.12	<u>70.13</u>	69.77	75.87	46.86	72.26
Ours (9B, fine-tuned)	<u>57.70</u>	44.79	44.80	96.12	70.58	<u>68.70</u>	<u>73.95</u>	<u>47.06</u>	72.99

backbones can retain competitive workflow-planning ability.

Observation 1: Earth-Agent-Pro improves cross-benchmark workflow fidelity. Under the shared GPT-5 backbone and evaluation protocol, Earth-Agent-Pro achieves the strongest Perception, GIS, and all three tool-order scores. It also improves answering accuracy by 11.16 points over the strongest GPT-5 baseline, indicating that the framework advantage transfers beyond Earth-Bench-Pro.

Observation 2: compact fine-tuned models retain strong procedural capabilities. The fine-tuned 9B model leads on GIS and AnyOrder, while the 4B model achieves the best SameOrder and Unique scores. These results suggest that role-specialized training enables small backbones to learn transferable workflow patterns across distinct benchmark settings.

c) *General-Purpose Agent Comparison:* We compare Earth-Agent-Pro with Codex and Manus across Spectrum, Products, and RGB tasks. Table V reports both task accuracy and mean wall-clock latency, exposing the trade-off between accuracy and execution cost.

Observation 1: a domain-specific workflow improves overall completion. Earth-Agent-Pro (GPT-5) obtains 75.00% average accuracy, 28.33 points above Codex, while also reducing average latency from 268.73 s to 208.64 s. Its largest

advantage appears on RGB tasks, where it completes all evaluated examples. The simultaneous gains in accuracy and latency indicate that improved task completion does not necessarily require higher execution cost.

d) *Framework Comparison on Earth-Bench-Pro:* We now directly isolate the effect of the Earth-Agent-Pro framework in a strictly controlled setting by comparing it with ReAct, AFlow, and OpenEarthAgent under the same GPT-5 backbone. Table VI provides evidence for the framework’s effectiveness on our target benchmark.

Observation 1: Earth-Agent-Pro leads seven of eight framework-comparison metrics. It improves LLM-as-Judge accuracy by 20.95 points over the strongest baseline and raises TAO from 70.47% to 87.61%. The corresponding gains in TIO, TEM, and parameter accuracy show that the improvement extends beyond selecting a plausible tool set to preserving more of the expert workflow structure.

D. Ablation Studies

a) *Domain-Skill Ablation:* We remove the retrieved domain skill while keeping the framework and model fixed within each pair. Table VII reports the result for ReAct,

TABLE V

COMPARISON WITH GENERAL-PURPOSE AGENTS. ACCURACY VALUES ARE REPORTED AS PERCENTAGES, AND AVERAGE LATENCY IS MEASURED IN SECONDS. BEST RESULTS ARE BOLDDED AND RUNNER-UPS ARE UNDERLINED.

Framework	Accuracy (%)				Avg. Latency (s)
	Spectrum	Products	RGB	Avg.	
Codex	40.00	75.00	25.00	46.67	268.73
Manus	40.00	45.00	45.00	43.33	291.00
Earth-Agent-Pro (GPT-5)	70.00	<u>55.00</u>	100.00	75.00	208.64
Earth-Agent-Pro (Qwen3.5-9B)	<u>45.00</u>	40.00	<u>65.00</u>	50.00	67.83
Earth-Agent-Pro (Qwen3.5-9B, fine-tuned)	70.00	40.00	<u>65.00</u>	<u>58.33</u>	<u>111.79</u>

TABLE VI

FRAMEWORK COMPARISON ON EARTH-BENCH-PRO USING A SHARED GPT-5 BACKBONE. ALL QUALITY METRICS ARE REPORTED AS PERCENTAGES. EFFICIENCY IS MEASURED AS A RATIO, WHERE VALUES CLOSER TO 1 INDICATE BETTER PERFORMANCE. BEST RESULTS ARE BOLDDED AND RUNNER-UPS ARE UNDERLINED.

Framework	LLM-as-Judge	ROUGE-L	EM	Efficiency	TAO	TIO	TEM	Param.
ReAct	<u>45.18</u>	<u>19.75</u>	<u>26.60</u>	<u>0.8636</u>	57.83	<u>47.53</u>	<u>43.33</u>	<u>24.33</u>
AFlow	40.16	10.80	36.70	0.8511	<u>70.47</u>	45.46	37.18	20.33
OpenEarthAgent	41.53	13.83	25.00	0.8101	60.06	42.62	39.32	22.67
Earth-Agent-Pro	66.13	21.68	13.31	1.0911	87.61	71.97	60.14	27.28

TABLE VII

ABLATION STUDY OF DOMAIN SKILLS. ALL QUALITY METRICS ARE REPORTED AS PERCENTAGES, WHILE EFFICIENCY IS MEASURED AS A RATIO. BOLD VALUES INDICATE THE BETTER RESULT WITHIN EACH ADJACENT PAIR.

Framework	Model	Skills	LLM-as-Judge	ROUGE-L	EM	Efficiency	TAO	TIO	TEM	Param.
ReAct	Qwen3.5-9B	×	34.27	13.97	10.08	2.1549	56.46	31.39	21.55	10.44
		✓	39.92	14.00	11.69	2.2145	57.13	37.66	29.76	12.14
Earth-Agent-Pro	Qwen3.5-9B	×	35.08	15.99	15.32	0.9955	75.61	61.94	51.71	24.84
		✓	38.31	16.33	15.73	1.0129	85.59	74.57	67.29	25.05
	Ours (9B, fine-tuned)	×	37.10	15.11	14.92	1.1405	82.41	70.93	55.92	30.00
		✓	50.00	20.65	19.76	1.1571	91.28	81.11	71.01	32.77

Earth-Agent-Pro with the base 9B model, and Earth-Agent-Pro with the fine-tuned 9B model. Bold values indicate the better result within each with/without-skill pair; for Efficiency, better means closer to 1.

Observation 1: domain skills improve reported quality metric. The improvement is particularly strong for trajectory fidelity. For the base Earth-Agent-Pro model, adding skills raises TIO from 61.94% to 74.57% and TEM from 51.71% to 67.29%. This pattern is consistent with skills constraining the workflow search space toward expert processing procedures.

Observation 2: skills and role-specialized training are complementary. For the fine-tuned model, adding skills improves LLM-as-Judge accuracy by 12.90 points, ROUGE-L by 5.54 points, and EM by 4.84 points. It also raises TAO, TIO, TEM, and parameter accuracy by 8.87, 10.18, 15.09, and 2.77 points, respectively. These simultaneous gains across all metrics show that the external skill prior remains effective after role-specialized training. The two components operate

at different levels: training strengthens the learned planning and execution policies, whereas skills provide explicit domain procedures that constrain workflow construction. Their combination consequently achieves the strongest answer-quality and trajectory-fidelity results, confirming that training does not make external skill guidance redundant.

b) Role-Specialized Training Ablation: Finally, we disentangle the effects of Planner SFT and Executor GRPO through adapter composition, planning-only evaluation, and oracle-plan execution under carefully controlled component substitutions. Base and Tuned denote the pretrained and role-specialized variants, respectively.

Adapter composition. Table VIII evaluates all four combinations of the Planner and Executor adapters. Bold values indicate the best result and underlined values indicate the runner-up; for Efficiency, values closer to 1 are better.

Observation 1: Planner and Executor tuning provide complementary benefits. Planner tuning produces the larger

TABLE VIII

ADAPTER COMPOSITION FOR ROLE-SPECIALIZED TRAINING ON EARTH-BENCH-PRO. ALL QUALITY METRICS ARE PERCENTAGES, WHILE EFFICIENCY IS A RATIO.

Planner	Executor	LLM-as-Judge	ROUGE-L	EM	Efficiency	TAO	TIO	TEM	Param.
Base	Base	38.31	16.33	15.73	1.0129	85.59	74.57	67.29	25.05
Base	Tuned	43.15	17.31	18.15	<u>0.9865</u>	85.44	72.24	65.64	28.34
Tuned	Base	<u>44.76</u>	20.67	<u>18.55</u>	1.1839	<u>89.47</u>	<u>78.78</u>	<u>70.09</u>	<u>30.33</u>
Tuned	Tuned	50.00	<u>20.65</u>	19.76	1.1571	91.28	81.11	71.01	32.77

TABLE IX

PLANNING-ONLY EVALUATION ON EARTH-BENCH-PRO. BOLD VALUES INDICATE THE BETTER RESULT WITHIN THE BASE/TUNED PAIR.

Planner	TAO	TIO	TEM
GPT-5	88.86	76.38	70.03
InternLM3-8B	72.15	55.38	46.87
GEMMA-4 (E4B-it)	83.75	71.34	61.80
MiniMax-M2.7	87.90	71.89	67.66
Base	86.70	71.87	66.96
Tuned	92.51	83.35	76.14

TABLE X

ORACLE-PLAN EXECUTOR EVALUATION ON EARTH-BENCH-PRO. EFF. DENOTES EFFICIENCY. BOLD VALUES INDICATE THE BETTER RESULT WITHIN THE BASE/TUNED PAIR.

Executor	LLM-as-Judge	ROUGE-L	EM	Eff.	Param.
GPT-5	68.95	24.31	18.15	0.9872	39.58
InternLM3-8B	38.71	12.04	10.48	0.8620	31.13
GEMMA-4 (E4B-it)	33.87	12.74	10.89	0.7109	33.66
MiniMax-M2.7	51.21	19.44	18.15	0.8309	35.72
Base	46.37	21.92	20.16	0.9081	34.19
Tuned	52.82	25.92	23.39	0.9274	37.20

gains in workflow metrics, whereas Executor tuning primarily improves final-answer quality and parameter grounding. Their combination achieves the strongest overall Judge, EM, TAO, TIO, TEM, and parameter scores.

Observation 2: end-to-end parameter accuracy also depends on planning. The tuned-Planner/base-Executor configuration has higher parameter accuracy than the base-Planner/tuned-Executor configuration. This occurs because the end-to-end metric also depends on workflow alignment and execution progress, motivating the isolated evaluations below.

Planning-only isolates workflow composition.

The planning-only evaluation isolates the Planner’s prediction of workflow structure. We run only the four-role sequential debate and compare its output tool sequence directly with the reference workflow. This setting omits tool execution and Executor-side tool replacement, early termination, same-node retries, suffix replanning, and workflow editing. The resulting metrics measure static workflow composition because the planning-only trace retains proposed tools without

execution, while the complete system records only verified tool calls. Planning-only scores can therefore exceed end-to-end trajectory scores when execution failures shorten the accepted trace.

Observation 3: Planner SFT improves workflow composition. The tuned Planner reaches 92.51% TAO, 83.35% TIO, and 76.14% TEM, improving over the base Planner by 5.81, 11.48, and 9.18 percentage points, respectively. It also exceeds GPT-5 by 3.65, 6.97, and 6.11 points on the same metrics. These gains isolate improved prediction of the reference tool set and order; end-to-end completion remains subject to subsequent execution.

Oracle-plan execution isolates argument grounding.

The oracle-plan evaluation isolates the Executor by fixing the workflow structure. We replace the predicted plan with the reference tool order and disable tool replacement, early termination, suffix replanning, and workflow editing. Each node still generates arguments, invokes the specified tool, verifies the returned observation, and retries the same tool when needed. This protocol fixes tool selection and ordering while preserving runtime argument grounding and environment interaction throughout the complete execution process.

Observation 4: Executor GRPO improves argument grounding. Relative to the base Executor, the tuned Executor raises LLM-as-Judge accuracy from 46.37% to 52.82%, ROUGE-L from 21.92% to 25.92%, EM from 20.16% to 23.39%, and parameter accuracy from 34.19% to 37.20%. Efficiency also moves from 0.9081 to 0.9274, closer to the ideal value of 1. GPT-5 still reaches 68.95% LLM-as-Judge accuracy and 39.58% parameter accuracy, indicating a remaining gap associated with backbone capability.

Together with the adapter-composition results in Table VIII, these isolated evaluations show that Planner SFT improves workflow composition and Executor GRPO improves argument grounding under a fixed workflow. The combined system benefits from these distinct interface-specific gains.

VII. DISCUSSION

A. Qualitative Case Studies

We further analyze three representative execution traces covering spectral observations, RGB imagery, and remote-sensing products, as shown in Figure 7. These examples complement the aggregate results by illustrating three factors that determine successful open-world EO execution: workflow completeness, preservation of intermediate dependencies, and correct grounding of tools and arguments.

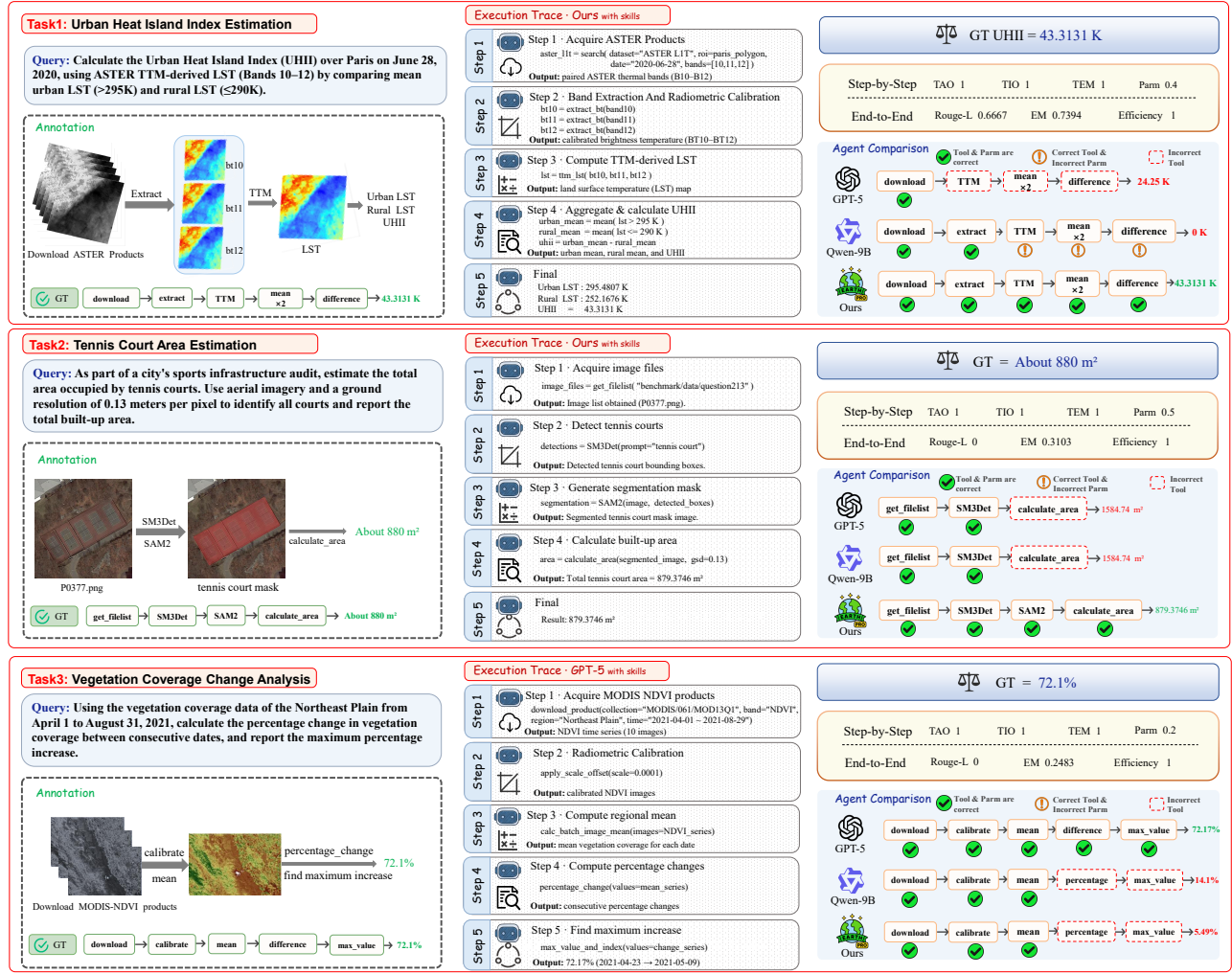


Fig. 7. **Representative execution cases on Earth-Bench-Pro.** From top to bottom: Spectrum-based UHII estimation, RGB-based tennis-court area estimation, and Product-based vegetation-coverage change analysis. The first two cases show that our agent preserves the required tool dependencies and matches the reference answers, whereas the third illustrates errors caused by incorrect downstream tool selection.

In the Spectrum case, the agent must acquire ASTER products, extract Bands 10–12, retrieve land-surface temperature using TTM, aggregate the urban and rural temperatures, and calculate their difference. GPT-5 skips the band extraction stage and applies the operations to incorrectly prepared inputs, producing 24.25 K. Qwen-9B includes extraction but incorrectly grounds the thermal-processing and aggregation operations, resulting in 0 K. In contrast, our agent preserves the executable dependency chain and recovers the reference UHII of 43.3131 K. This case shows that selecting plausible tools is insufficient when their inputs and arguments are not grounded in the preceding observations.

The RGB case further demonstrates the importance of intermediate representations. GPT-5 and Qwen-9B detect the tennis courts but proceed directly from the detections to area calculation. Without converting the coarse detections into a pixel-level mask, both agents overestimate the occupied area as 1584.74 m². Our agent inserts SAM2 segmentation between detection and measurement and calculates the area from the resulting mask using the specified ground resolution. It therefore obtains 879.3746 m², consistent with the reference answer

of approximately 880 m². The difference arises not from the final measurement tool alone, but from whether the correct intermediate artifact is constructed and passed downstream.

The Product case exposes a limitation. All three agents perform MODIS acquisition, radiometric calibration, and regional-mean calculation. GPT-5 follows the reference change-analysis and maximum-value operations, obtaining 72.17%, close to the reference value of 72.1%. Qwen-9B and our agent instead select incorrect downstream operations, producing 14.1% and 5.49%, respectively. Thus, correct upstream data preparation does not guarantee a correct answer when the scientific meaning of a downstream operation is mismatched.

Taken together, these cases show that open-world EO execution requires more than individually reasonable tool calls. Reliable quantitative answers depend on maintaining the complete workflow, constructing the intended intermediate representations, and grounding every downstream operation in both the accepted execution evidence and the scientific objective. The Product failure also indicates that skill guidance improves workflow reliability but does not fully eliminate semantic tool-selection errors.

B. Limitations and Trade-offs

Open-world EO execution remains limited by exact workflow fidelity and numerical grounding. Even the strongest TEM and parameter scores in Table II are 71.01% and 32.77%, respectively. The gap between reliable tool-set coverage and much lower parameter accuracy suggests that selecting relevant tools is easier than instantiating a fully faithful executable workflow. Open-ended QA removes answer candidates and requires the model or agent to directly produce a calibrated numerical value, which more closely reflects real EO requests for measurements. It also requires target localization, metric reasoning, unit handling, and correct answer formatting. The zero and near-zero Distance accuracy across most methods in Table III shows that precise open-ended spatial measurement remains an unresolved challenge for both remote-sensing MLLMs and current agent systems.

Workflow fidelity and final-task accuracy remain distinct, and capability profiles vary across models and modalities. In Table IV, AFlow leads Operation F1 and ReAct leads image-generation accuracy. The fine-tuned 4B and 9B models achieve higher tool-order fidelity than Earth-Agent-Pro (GPT-5) but remain below it in answering accuracy, suggesting that final-answer synthesis remains more dependent on backbone capability. On Product tasks in Table V, Codex reaches 75.00% accuracy, compared with 55.00% for the strongest Earth-Agent-Agent-Pro variant. This modality-specific result limits any claim of uniform superiority across all task families.

Accuracy improvements also incur additional execution cost. Table V shows that fine-tuning the 9B variant raises average accuracy from 50.00% to 58.33% and latency from 67.83 s to 111.79 s. The fine-tuned small model remains faster than Earth-Agent-Pro (GPT-5), whose latency is 208.64 s, while closing part of the accuracy gap. In Table VI, Earth-Agent-Pro achieves an Efficiency ratio of 1.0911, whose distance to the ideal value of 1 is smaller than that of ReAct, AFlow, or OpenEarthAgent. Its trajectories are slightly longer than the reference on average, whereas the baselines tend to terminate with fewer calls. Table VII further shows that adding skills moves Efficiency farther from 1 in all three ablations, increasing the absolute deviation by 0.0596 for ReAct, 0.0084 for the base Earth-Agent-Pro model, and 0.0166 for the fine-tuned model. This modest length overhead accompanies consistent improvements in answer quality and trajectory fidelity and may reflect additional domain-recommended processing or verification steps, exposing a quality-length trade-off.

C. Error Analysis

Figure 8 shows how training, skill guidance, and framework design affect operational reliability. In Panel A1, role-specific training reduces file-hallucination events from nine to seven, while our fine-tuned Qwen3.5-9B model matches GPT-5 at nine logged invalid-parameter events. Panel A2 shows the backbone gap: Mistral-7B and Llama-3.2-3B retain 138 and 280 errors under WS, dominated by tool hallucinations and invalid parameters. Panel B shows that skills eliminate logged tool-hallucination events in both Qwen3.5-9B configurations and reduce total errors by 36.0% for the fine-tuned model and

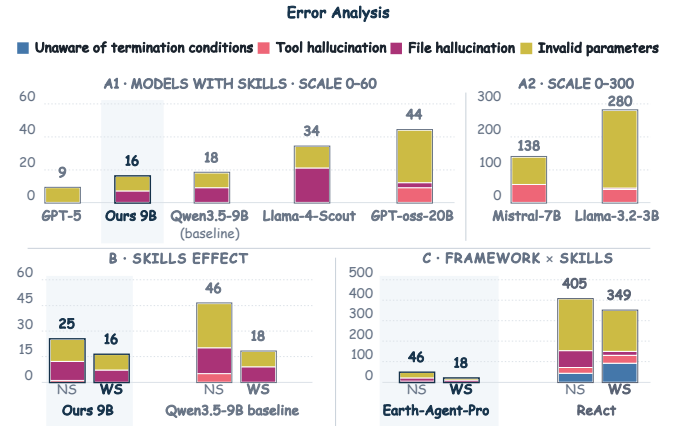


Fig. 8. **Skills reduce operational error counts.** Each run contains 248 trajectories. Bars count logged operational errors, and one trajectory may contribute multiple events. Panel A compares with-skills (WS) counts across models, while Panel B compares no-skills (NS) with WS for the two 9B configurations. Panel C compares Earth-Agent-Pro and ReAct under the shared Qwen3.5-9B backbone. Colors denote repeated no-progress calls until the step limit, calls to nonexistent tools, accesses to nonexistent files or paths, and parameter schema, format, or value violations.

60.9% for the base model. The larger base-model gain shows that skills transfer across configurations and remain effective without role-specific training. Panel C holds Qwen3.5-9B fixed and shows that Earth-Agent-Pro records fewer events than ReAct in every error category under NS and WS, lowering the total by 88.6% and 94.8%, respectively.

VIII. CONCLUSION

This work studies open-world EO execution, where an agent must locate runtime inputs, acquire observations when required, prepare data, perform domain computations, and generate an open-ended answer from runtime evidence given a high-level scientific question. Earth-Bench-Pro instantiates 248 expert-curated task cores as 744 matched questions, supporting comparisons across instruction following, autonomous planning, and open-world execution through executable evidence chains. Earth-Agent-Pro connects workflow planning with runtime operation grounding through expert skill guidance, workflow-centered structured memory, and localized suffix repair. Role-specialized training uses sequence-level SFT for Planner workflow composition and node-level GRPO for Executor argument grounding. On Earth-Bench-OW, Earth-Agent-Pro leads seven of eight framework-comparison metrics under a shared GPT-5 backbone and reaches 66.13% LLM-as-Judge accuracy. Adapter composition and isolated evaluations record higher workflow-composition scores for the tuned Planner and higher answer and parameter scores for the tuned Executor under a fixed workflow.

The experiments also expose remaining constraints for open-world EO agents. Even the strongest configurations achieve higher tool coverage than exact workflow matching and parameter accuracy, while final-answer quality remains sensitive to backbone capability. Future work should improve argument grounding under runtime uncertainty, reduce the trajectory overhead introduced by skill-guided execution, and strengthen evidence synthesis in compact models.

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